Type MS · Solving an equation with your graph and a straight line

Answer 1 of 12 · Question 8 part c · 5 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

8 (a)

x	0.5	0.75	1	1.5	2	3	4	5
y	10	**3.83**	2	**1.33**	**1.75**	**3.33**	5.19	7.12

B2 [2]

(b)

Graph drawn: All points to within half of a square — B1ft
All points joined together in a smooth curve — B1ft
[2]

(c)

$2x+\frac{3}{x^2}-3=ax+b \Rightarrow 2x^3+3-3x^2=ax^3+bx^2$

$\Rightarrow x^3(2-a)-x^2(3+b)+3\equiv 4x^3-10x^2+3$ — M1

$2-a=4 \Rightarrow a=-2$

$3+b=10 \Rightarrow b=7$ — M1A1

So line is $y=7-2x$

$x=0.6,\ 2.4$ — M1A1

[Accept $x=0.7$ and $x=2.3$] [5]

ALT

$\frac{4x^3-10x^2+3}{x^2}=0 \Rightarrow 4x-10+\frac{3}{x^2}=0$ — {M1

$2x+\frac{3}{x^2}=10-2x \Rightarrow 2x+\frac{3}{x^2}-3=7-2x$ so line is $y=7-2x$ — M1A1

$x=0.6,\ 2.4$

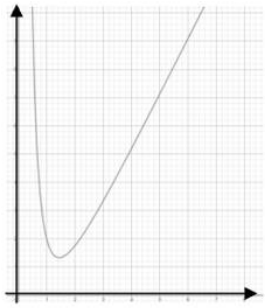
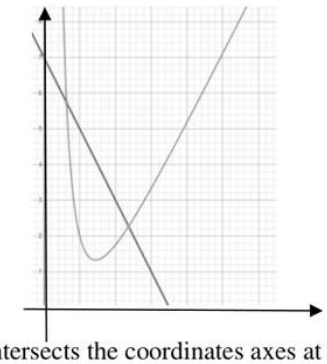
[Accept $x=0.7$ and $x=2.3$] — M1A1 [5]}

Total 9 marks

Type MS · Solving an equation with your graph and a straight line

Answer 1 of 12 · Question 8 part c · 5 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	For two out of 4 values rounded correctly. Penalise rounding only once for awrt the required values.
	B1	For all four values correctly rounded
(b)	B1ft	For **their** points plotted correctly to within half of a square.
	B1ft	For their points joined up in a smooth curve. 
(c)	M1	For setting $2x+\frac{3}{x^2}-3=ax+b$ and attempting to simplify to the form $x^3(2-a)-x^2(3+b)+3$ where the coefficient of each of x^3 and x^2 is in terms of a and/or b
	M1	For equating coefficients and attempting to find the value of a and b
	ALT	
	M1	For dividing through by x^2 to give $4x-10+\frac{3}{x^2}=0$
	M1	For adding 7 and subtracting $2x$ from both sides to give on the LHS $2x+\frac{3}{x^2}-3$
	A1	For the correct straight line $y=7-2x$
	M1	For their line drawn provided it is of the form $y=A-2x$ where $A\neq 0$  The line intersects the coordinates axes at (0, 7) and (3.5, 0)
	A1	For both values of 0.6 and 2.4 Accept 0.7 and 2.3 - without working these roots score no marks. **NB**: Calculator roots are 2.366 and 0.634

Type MS · Solving an equation with your graph and a straight line

Answer 1 of 12 · Question 8 part c · 5 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Solving an equation with your graph and a straight line

Answer 2 of 12 · Question 8 part c · 3 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Number		
8 (a)	<table><tr><td>x</td><td>0</td><td>0.25</td><td>0.5</td><td>1</td><td>1.5</td><td>2</td><td>3</td></tr><tr><td>y</td><td>2</td><td>**2.41**</td><td>**2.69**</td><td>3.10</td><td>3.39</td><td>3.61</td><td>**3.95**</td></tr></table>	B2 [2]
(b)	Points plotted within half of a square Points joined together in a smooth curve	B1ft B1ft [2]
(c)	$\ln(2x+1)=3x-4 \Rightarrow \ln(2x+1)+2=3x-2$ Graph of $y=3x-2$ drawn. Intersection point is at $x=1.8$ or 1.9 (Accept either)	M1 M1A1 [3]
(d)	$e^{(6-x)}=(2x+1)^2 \Rightarrow 6-x=\ln(2x+1)^2 \Rightarrow 6-x=2\ln(2x+1)$ $\Rightarrow \ln(2x+1)+2=5-\frac{x}{2}$ Graph of $y=5-\frac{x}{2}$ drawn. Intersection point is at $x=2.4$ or 2.5 (Accept either)	M1 M1 M1A1 [4]
		Total 11 marks

Type MS · Solving an equation with your graph and a straight line

Answer 2 of 12 · Question 8 part c · 3 marks

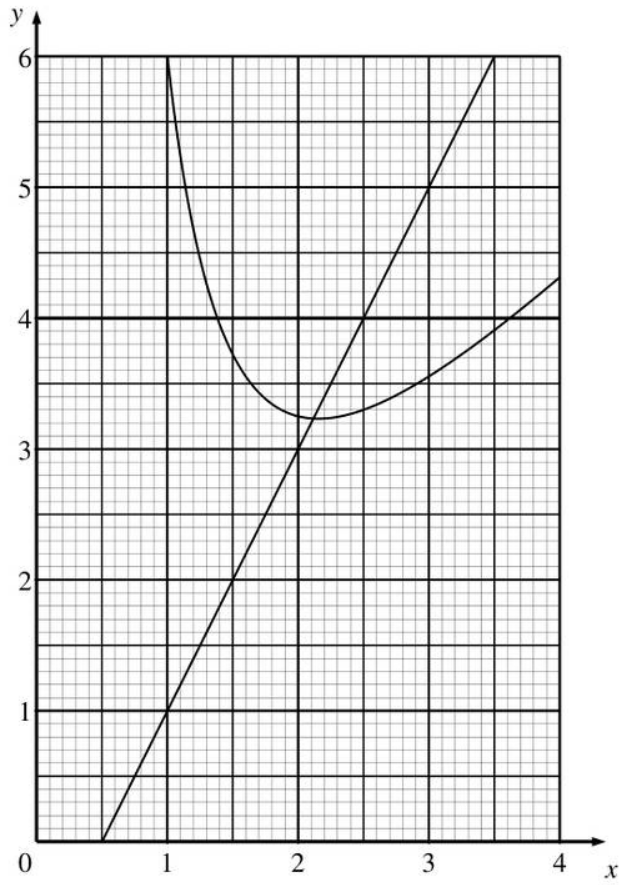
2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Solving an equation with your graph and a straight line

Answer 3 of 12 · Question 4 part a · 3 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

Number		
4		
(a)	$x^3-4x^2+5=0 \Rightarrow x-4+\frac{5}{x^2}=0$	
	$x+\frac{5}{x^2}=4$	M1
	Draw "$y=4$", $x=1.4,\ 3.6$	M1, A1 (3)
(b)	$x^3-x^2-5=0 \Rightarrow x-1-\frac{5}{x^2}=0$	
	$x-1-\frac{5}{x^2}=0 \Rightarrow 2x-1=x+\frac{5}{x^2}$	M1A1
	Draw "$y=2x-1$", $x=2.1$	dM1A1 (4) [7]

Type MS · Solving an equation with your graph and a straight line

Answer 3 of 12 · Question 4 part a · 3 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	Divides through $x^3-4x^2+5=0$ by x^2 and rearranges to achieve as a minimum $x+\frac{5}{x^2}=k$ where k is a constant $\left[x-4+\frac{5}{x^2}=0\Rightarrow x+\frac{5}{x^2}=4\right]$
	M1	**Draw the line $y=k$** following through their value for k No line is M0
	A1	For the two values of $x=1.4$ and $x=3.6$ Condone answers given as coordinates provided they are completely correct. (1.4, 4) and (3.6, 4) Require both M marks for this mark.
(b)	M1	For setting $x+\frac{5}{x^2}=Ax+B\Rightarrow x^3+5=Ax^3+Bx^2\Rightarrow Ax^3-x^3+Bx^2-5=0$, and equating coefficients with x^3-x^2-5 $x^3(A-1)+Bx^2-5\equiv x^3-x^2-5$ to achieve as a minimum $A=(\pm 2)$, $B=(\pm 1)$
	A1	For the correct straight line $y=2x-1$
	ALT – to find the line $y=2x-1$	
	M1	Divides through $x^3-x^2-5=0$ by x^2 and rearranges the equation to achieve as a minimum $\Rightarrow \pm 2x\pm 1=x+\frac{5}{x^2}$
	A1	For the correct straight line $y=2x-1$
	dM1	**Draws their $y=2x-1$ on the graph** and locates the point of intersection. Please check that they draw their line correctly. Coordinates for you to check are (0.5, 0) and (2.5, 4) No line is M0 This mark is dependent on the first M mark in (b)
	A1	For the correct value of $x=2.1$ [allow $x=2.2$] Can only score this mark from M1A1M1 Do not accept the answer given as coordinates.

Type MS · Solving an equation with your graph and a straight line

Answer 3 of 12 · Question 4 part a · 3 marks

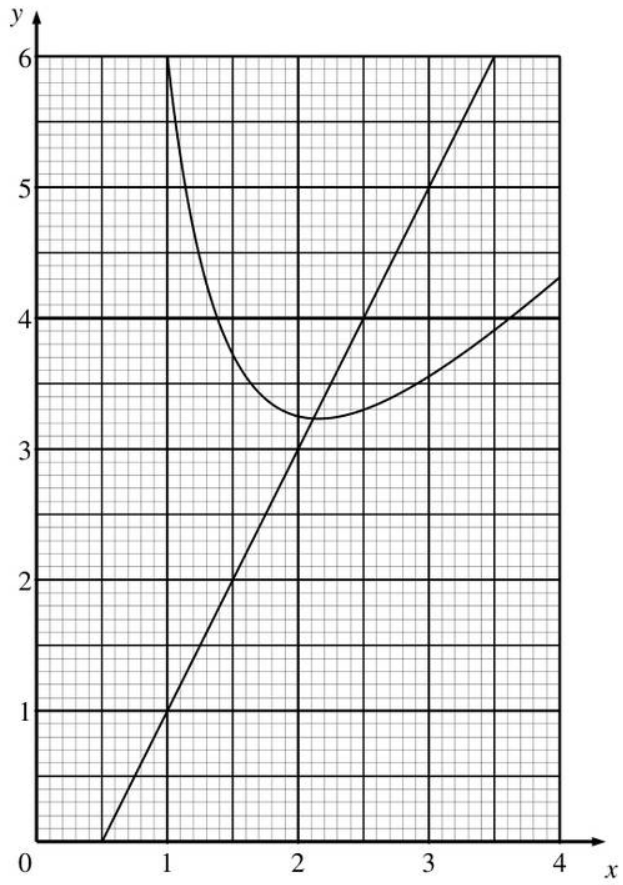
2020 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	6 8	

Type MS · Solving an equation with your graph and a straight line

Answer 4 of 12 · Question 4 part b · 4 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

Number		
4		
(a)	$x^3 - 4x^2 + 5 = 0 \Rightarrow x - 4 + \frac{5}{x^2} = 0$	
	$x + \frac{5}{x^2} = 4$	M1
	Draw "$y = 4$", $x = 1.4,\ 3.6$	M1, A1 (3)
(b)	$x^3 - x^2 - 5 = 0 \Rightarrow x - 1 - \frac{5}{x^2} = 0$	
	$x - 1 - \frac{5}{x^2} = 0 \Rightarrow 2x - 1 = x + \frac{5}{x^2}$	M1A1
	Draw "$y = 2x - 1$", $x = 2.1$	dM1A1 (4)
		[7]

Type MS · Solving an equation with your graph and a straight line

Answer 4 of 12 · Question 4 part b · 4 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	Divides through $x^3-4x^2+5=0$ by x^2 and rearranges to achieve as a minimum $x+\frac{5}{x^2}=k$ where k is a constant $\left[x-4+\frac{5}{x^2}=0\Rightarrow x+\frac{5}{x^2}=4\right]$
	M1	**Draw the line $y=k$** following through their value for k No line is M0
	A1	For the two values of $x=1.4$ and $x=3.6$ Condone answers given as coordinates provided they are completely correct. (1.4, 4) and (3.6, 4) Require both M marks for this mark.
(b)	M1	For setting $x+\frac{5}{x^2}=Ax+B\Rightarrow x^3+5=Ax^3+Bx^2\Rightarrow Ax^3-x^3+Bx^2-5=0$, and equating coefficients with x^3-x^2-5 $x^3(A-1)+Bx^2-5\equiv x^3-x^2-5$ to achieve as a minimum $A=(\pm 2)$, $B=(\pm 1)$
	A1	For the correct straight line $y=2x-1$
	ALT – to find the line $y=2x-1$	
	M1	Divides through $x^3-x^2-5=0$ by x^2 and rearranges the equation to achieve as a minimum $\Rightarrow \pm 2x\pm 1=x+\frac{5}{x^2}$
	A1	For the correct straight line $y=2x-1$
	dM1	**Draws their $y=2x-1$ on the graph** and locates the point of intersection. Please check that they draw their line correctly. Coordinates for you to check are (0.5, 0) and (2.5, 4) No line is M0 This mark is dependent on the first M mark in (b)
	A1	For the correct value of $x=2.1$ [allow $x=2.2$] Can only score this mark from M1A1M1 Do not accept the answer given as coordinates.

Type MS · Solving an equation with your graph and a straight line

Answer 4 of 12 · Question 4 part b · 4 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	6 8	

Type MS · Solving an equation with your graph and a straight line

Answer 5 of 12 · Question 4 part c · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

4 (a)	x: 0.5, 1, 1.5, 2, 2.5, 3, 3.5 y: 10, 5, 4.89, 5.5, 6.32, 7.22, 8.16	B2 (2)
(b)	Drawn	B1ft B1ft (2)
(c)	$2x+1+\frac{2}{x^2}=8$ $x=0.6$ $x=3.4$	M1 A1 (2)
(d)	$2x+1+\frac{2}{x^2}=\frac{1}{2}x+6$ $y=\frac{1}{2}x+6$ drawn $x=0.7$ $x=3.2$	M1 A1 M1 A1 A1 (5)
		[11]

	Notes
(a)	
B2	For all 4 values correct (B1 for at least 2 values correct)
(b)	
B1ft	For points plotted ft their table (allow half square tolerance)
B1ft	For points joined together with a smooth curve ft their table
(c)	
M1	For $2x+1+\frac{2}{x^2}=8$ may be implied by [y =] 8 identified on the graph
A1	For 0.6 and 3.4 (Allow 0.7 for 0.6)
(d)	
M1	For adding $\frac{1}{2}x$ to both sides of the equation or adding 1 to both sides of the equation
A1	For $2x+1+\frac{2}{x^2}=\frac{1}{2}x+6$
M1	For $y=\frac{1}{2}x+6$ drawn
A1	For 0.7 (allow 0.8)
A1	For 3.2 (allow 3.3)

Type MS · Solving an equation with your graph and a straight line

Answer 5 of 12 · Question 4 part c · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Solving an equation with your graph and a straight line

Answer 6 of 12 · Question 4 part d · 5 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

4 (a)	x: 0.5, 1, 1.5, 2, 2.5, 3, 3.5 y: 10, 5, 4.89, 5.5, 6.32, 7.22, 8.16	B2 (2)
(b)	Drawn	B1ft B1ft (2)
(c)	$2x+1+\frac{2}{x^2}=8$ $x=0.6 \quad x=3.4$	M1 A1 (2)
(d)	$2x+1+\frac{2}{x^2}=\frac{1}{2}x+6$ $y=\frac{1}{2}x+6$ drawn $x=0.7 \quad x=3.2$	M1 A1 M1 A1 A1 (5)
		[11]

	Notes
(a)	
B2	For all 4 values correct (B1 for at least 2 values correct)
(b)	
B1ft	For points plotted ft their table (allow half square tolerance)
B1ft	For points joined together with a smooth curve ft their table
(c)	
M1	For $2x+1+\frac{2}{x^2}=8$ may be implied by [y =] 8 identified on the graph
A1	For 0.6 and 3.4 (Allow 0.7 for 0.6)
(d)	
M1	For adding $\frac{1}{2}x$ to both sides of the equation or adding 1 to both sides of the equation
A1	For $2x+1+\frac{2}{x^2}=\frac{1}{2}x+6$
M1	For $y=\frac{1}{2}x+6$ drawn
A1	For 0.7 (allow 0.8)
A1	For 3.2 (allow 3.3)

Type MS · Solving an equation with your graph and a straight line

Answer 6 of 12 · Question 4 part d · 5 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Solving an equation with your graph and a straight line

Answer 7 of 12 · Question 8 part d · 5 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part d in this mark scheme extract

<table>
<tr><td>8(a)</td><td>
<table>
<tr><td>x</td><td>0</td><td>0.5</td><td>0.8</td><td>1</td><td>1.6</td><td>2</td><td>2.5</td><td>3</td></tr>
<tr><td>y</td><td>–3.5</td><td>–1.6</td><td>0.3</td><td>1.6</td><td>3.5</td><td>2.4</td><td>–0.8</td><td>–3.2</td></tr>
</table>
</td><td>B2</td></tr>
<tr><td>(b)</td><td>Their points plotted correctly on the graph
Points are joined to form a smooth curve</td><td>B1ft
B1ft
[2]</td></tr>
<tr><td>(c)</td><td>$\cos(A+B)=\cos A\cos B-\sin A\sin B \Rightarrow \cos 2A=\cos^2 A-\sin^2 A=1-\sin^2 A-\sin^2 A$
$\cos(2A)=1-2\sin^2 A$</td><td>M1

A1
cso
[2]</td></tr>
<tr><td>(d)</td><td>$2\sin x+6(1-\cos 2x)-x-5=0$
$\sin x-3\cos 2x-\frac{1}{2}=\frac{x}{2}\pm k$ where k is a constant $\Rightarrow y=\frac{x}{2}-1$
Draws line $y=\frac{x}{2}-1$
$x=0.6$ or $x=0.7$ AND $x=2.3$ or $x=2.4$</td><td>M1

dM1A1

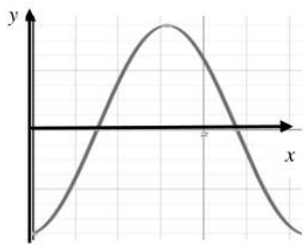
M1

A1
[5]</td></tr>
<tr><td colspan="2">Total</td><td>11 marks</td></tr>
</table>

Type MS · Solving an equation with your graph and a straight line

Answer 7 of 12 · Question 8 part d · 5 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part d in this mark scheme extract

<table>
<tr><th>Part</th><th>Mark</th><th>Scheme</th></tr>
<tr><td>(a)</td><td>B2</td><td>
<table>
<tr><td>x</td><td>0</td><td>0.5</td><td>0.8</td><td>1</td><td>1.6</td><td>2</td><td>2.5</td><td>3</td></tr>
<tr><td>y</td><td>–3.5</td><td>–1.6</td><td>0.3</td><td>1.6</td><td>3.5</td><td>2.4</td><td>–0.8</td><td>–3.2</td></tr>
</table>
For all 4 values correct OR If B2 not scored then award B1 for any 2 values correct
Note: These values must be rounded to one decimal place only. Accept only the above values.
</td></tr>
<tr><td>(b)</td><td>B1ft
B1ft</td><td>


All points plotted correctly (ft their values) on the graph within half of one square.
All of their points are joined to form a smooth curve within half of a square of a point. There must be at least six points plotted and joined.
</td></tr>
<tr><td rowspan="2">(c)</td><td>M1</td><td>For using the identity $\cos(A+B)=\cos A\cos B-\sin A\sin B=\cos^2 A-\sin^2 A$ correctly and the identity $\sin^2 A+\cos^2 A=1$ also correctly.</td></tr>
<tr><td>A1
cso</td><td>For simplifying to the required identity $\cos 2A=1-2\sin^2 A$ *
This is a given result so there must be no errors in their solution.</td></tr>
<tr><td rowspan="5">(d)</td><td>M1</td><td>$2\sin^2 x=1-\cos 2x$
For substituting the above into the given f (x) with no errors.
e.g., $2\sin x+6(1-\cos 2x)-x-5=0$</td></tr>
<tr><td>dM1</td><td>For reaching the equation of the curve on one side and the equation of a straight line on the other.
$\sin x-3\cos 2x-\frac{1}{2}=\frac{x}{2}\pm k$ where k is a constant o.e
Note: This mark is dependent on the previous M mark.</td></tr>
<tr><td>A1</td><td>For the correct equation of the equation of the straight line required $y=\frac{x}{2}-1$</td></tr>
<tr><td>M1</td><td>For drawing their line which must be in the form $y=\frac{x}{2}\pm k$
Correct coordinates for $y=\frac{x}{2}-1$ are; $(0,\ -1),\ \left(1,\ -\frac{1}{2}\right),\ (2,\ 0),\ \left(3,\ \frac{1}{2}\right)$</td></tr>
<tr><td>A1</td><td>For both correct roots of $x=0.6$ or $x=0.7$ AND $x=2.3$ or $x=2.4$</td></tr>
</table>

Type MS · Solving an equation with your graph and a straight line

Answer 7 of 12 · Question 8 part d · 5 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part d in this mark scheme extract

Question	Scheme	Mark

Type MS · Solving an equation with your graph and a straight line

Answer 8 of 12 · Question 6 part c · 5 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

6 (a)	<table><tr><td>x</td><td>0.5</td><td>1</td><td>1.5</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>−11.5</td><td>**-2**</td><td>**0.2**</td><td>1.3</td><td>2.7</td><td>**3.8**</td><td>**4.9**</td><td>5.9</td></tr></table>	B2 [2]
(b)	Points plotted within half of a square Points joined together in a smooth curve	B1ft B1ft [2]
(c)	$\frac{x^3-3}{x^2}=ax+b \Rightarrow x^3-3=ax^3+bx^2 \Rightarrow 0=x^3(a-1)+bx^2+3$ Comparing coefficients $x^3(a-1)-bx^2+3=2x^3+6x^2+3$ $\Rightarrow a=3, b=-6$ so line required is $y=3x-6$ Draws the line $y=3x-6$ and identifies two intersections with the curve when $x=0.8/0.9$ and $x=2.8/2.9$	M1 M1A1 M1 A1 (both) [5]
	ALT	
	$2x^3-6x^2+3=0 \Rightarrow 2x-6=-\frac{3}{x^2}$	{M1}
	$3x-6=x-\frac{3}{x^2}$ so line required is $y=3x-6$	{M1}{A1}
	Draws the line $y=3x-6$ and identifies two intersections with the curve when $x=0.8/0.9$ and $x=2.8/2.9$	{M1} {A1} both [5]
		Total 9 marks

Type MS · Solving an equation with your graph and a straight line

Answer 8 of 12 · Question 6 part c · 5 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

(a)	
B2	All 4 points correct (**B1** for 3 points correct)
(b)	
B1ft	Points plotted ft their table allow half a square tolerance
B1ft	Points joined together with a smooth curve ft their table
(c)	
M1	Setting $x-\frac{3}{x^2}=ax+b$ and simplifying to $x^3(a-1)+bx^2+3$
M1	Comparing coefficients
A1	Identifying that the line required is $y=3x-6$
M1	$y=3x-6$ drawn intersecting the curve in two places
A1	$x=0.8/0.9$ and $x=2.8/2.9$
ALT	
M1	Subtracting 3 from both sides and dividing by x^2
M1	Adding x to both sides
A1	Identifying that the line required is $y=3x-6$
M1	$y=3x-6$ drawn intersecting the curve in two places
A1	$x=0.8/0.9$ and $x=2.8/2.9$

Type MS · Solving an equation with your graph and a straight line

Answer 8 of 12 · Question 6 part c · 5 marks

2020 JAN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Solving an equation with your graph and a straight line

Answer 9 of 12 · Question 6 part c · 3 marks

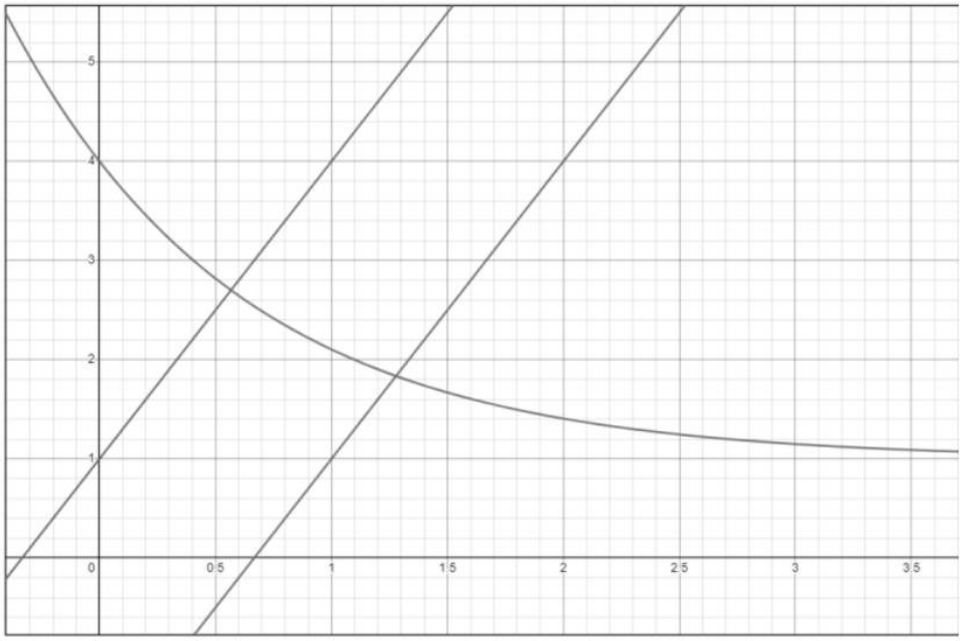
2022 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

6 a	**0** *0.25* *0.5* **1** *1.5* **2** *3* **4(.00)** *3.34* *2.82* **2.1(0)** *1.67* **1.41** *1.15*	B2 (2)
b	Points plotted within half a square Points joined with a smooth curve	B1 ft B1 ft (2)
c	$x = e^{-x} \Rightarrow 3x + 1 = 1 + 3e^{-x}$ or sight of $y = 3x + 1$ and $y = 1 + 3e^{-x}$ $y = 3x + 1$ drawn. Intersection is at $x = 0.5$ or 0.6	M1 M1 A1 (3)
d	$\ln(x-1)^3 = -3x \Rightarrow \ln(x-1) = -x$ $\Rightarrow 3x - 3 = 3e^{-x} \Rightarrow 3x - 2 = 1 + 3e^{-x}$ $y = 3x - 2$ drawn. Intersection is at $x = 1.2$ or 1.3	M1 M1 M1 A1 (4)
ALT – first 2 marks	$(x-1)^3 = e^{-3x} \Rightarrow \left(\sqrt[3]{(x-1)^3} = \sqrt[3]{e^{-3x}}\right) \Rightarrow x - 1 = e^{-x}$ $3x - 3 = 3e^{-x} \Rightarrow 3x - 2 = 1 + 3e^{-x}$	M1 M1
		Total 11 marks

Type MS · Solving an equation with your graph and a straight line

Answer 9 of 12 · Question 6 part c · 3 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract



Type MS · Solving an equation with your graph and a straight line

Answer 9 of 12 · Question 6 part c · 3 marks

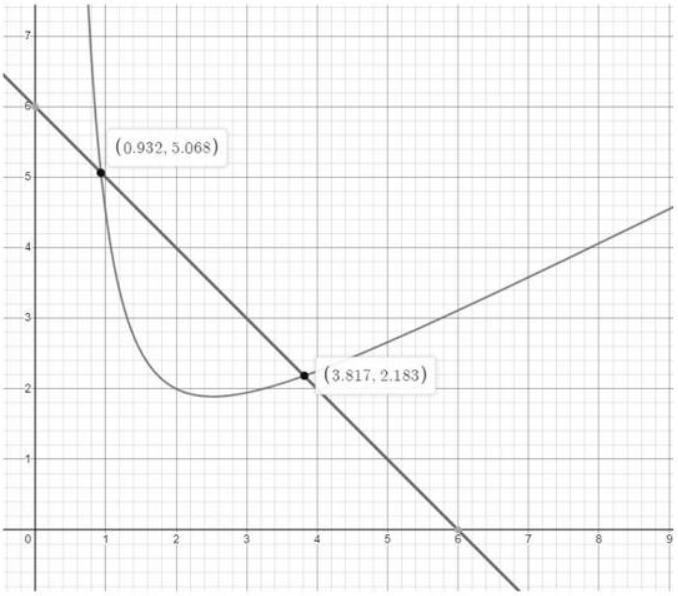
2022 JUN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks
	$f(x) = 4 \quad 12 \quad 19$	

Type MS · Solving an equation with your graph and a straight line

Answer 10 of 12 · Question 3 part whole · 5 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part whole in this mark scheme extract

3	$\frac{x}{2}+\frac{4}{x^2}=Ax+B \Rightarrow x^3(1-2A)-2Bx^2+8=0$ $x^3(1-2A)-2Bx^2+8=3x^3-12x^2+8 \Rightarrow A=-1,\ B=6 \Rightarrow y=6-x$ Draws line $y=6-x$ Roots are $x=0.9,\ 3.8$  	M1 M1A1 M1 A1 [5]
		Total 5 marks

Mark	Notes
M1	Removes the denominators by multiplying through by $2x^2$ This can be carried out separately by multiplying by 2 and then x^2 or vice versa. Allow one slip in their working. $\frac{x}{2}+\frac{4}{x^2}=Ax+B \Rightarrow x^3+8=2Ax^3+2Bx$
M1	Collects like terms and equates components. $x^3(1-2A)-2Bx^2+8=3x^3-12x^2+8$ and finds the equation of the straight line.
A1	The straight line is $y=6-x$
ALT 1	
M1	Divides through by $2x^2$ This can be carried out separately by dividing by 2 and then x^2 or vice versa.

Type MS · Solving an equation with your graph and a straight line

Answer 10 of 12 · Question 3 part whole · 5 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

	Allow one slip in their working. $3x^3-12x^2+8=0 \quad (\div 2x^2) \Rightarrow \frac{3x}{2}-6+\frac{4}{x^2}=0$
M1	Adds $(6-x)$ to both sides. $\Rightarrow \frac{x}{2}+\frac{4}{x^2}=6-x$
A1	The straight line is $y=6-x$
ALT 2	
M1	Multiplies through $\frac{x}{2}+\frac{4}{x^2}=y \Rightarrow x^3+8=2x^2y$ **and** subtracts this from $3x^3-12x^2+8=0$ as follows to obtain $2x^3-12x^2=-2x^2y$ $3x^3-12x^2+8=0$ $x^3 \quad +8=2x^2y$ $2x^3-12x^2 \quad =-2x^2y$
M1	Divides $2x^3-12x^2=-2x^2y$ by $2x^2$ to obtain a linear equation $x-6=-y$
A1	The straight line is $y=6-x$
SC: No or incorrect working to find the equation of the line. • If fully correct answers are seen without correct or any working at all and without the correct **equation** of the line seen [even if a line is drawn], award M0M0A0M0A0 • If the rounded answers to 1 dp are seen **with** the correct equation $y=6-x$, and the line drawn, award M0M0A0M1A1 • If rounded answers are seen to a greater degree of accuracy **with** the correct equation $y=6-x$ seen, and the line drawn, award M0M0A0M1A0	
M1	For drawing their straight line correctly onto the grid, provided it is of the form $y=k-x$
A1	For a value of 0.9 [allow 1.0] and 3.8 (do not award without the evidence of the line) Allow seen as coordinates (0.9, [5.1]) (3.8, [2.2]) ignore y values. Do not accept awrt in this question. Rounding must be to one decimal place.

Type MS · Solving an equation with your graph and a straight line

Answer 10 of 12 · Question 3 part whole · 5 marks

2023 JUN · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Solving an equation with your graph and a straight line

Answer 11 of 12 · Question 4 part whole · 4 marks

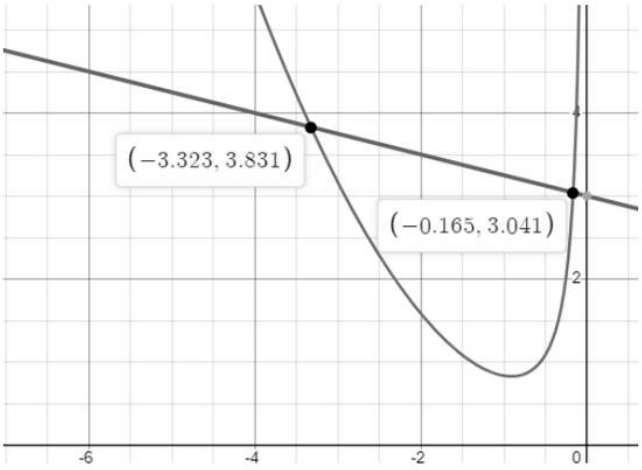
2024 JUN · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

4	$4x^3+3x^2-36x-6=0 \quad \rightarrow \div 12x$ $\Rightarrow \frac{x^2}{3}+\frac{x}{4}-3-\frac{1}{2x}=0 \Rightarrow \frac{x^2}{3}-\frac{1}{2x}=3-\frac{x}{4}$ **ALT** $\frac{x^2}{3}-\frac{1}{2x}=ax+b \Rightarrow 2x^3-6ax-6bx-3=0$ $\Rightarrow 4x^3-12ax-12bx-6=4x^3+3x-36x-6$ $\Rightarrow a=-\frac{1}{4} \quad b=3$ $\Rightarrow$ line required is $y=3-\frac{x}{4}$ Draws the line with equation $y=3-\frac{x}{4}$ $x=-0.2 \quad x=-3.3$ or -3.4	M1A1 [M1 A1] M1 A1 [4]
		Total 4 marks

Type MS · Solving an equation with your graph and a straight line

Answer 11 of 12 · Question 4 part whole · 4 marks

2024 JUN · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Notes
Note	• Correct values **without** any evidence of valid working or a line is M0A0M0A0 • Correct values **with** a correct line drawn without valid working is M0A0M0A0 • Correct values **with** a line drawn **and** the correct equation of a line without evidence of any valid working is SC M0A0M1A0
M1	For dividing through by $12x$ [you may well see this in stages – e.g., first by 4, then by 3, then by x etc] and attempting to rearrange the equation to give as a minimum $y = k \pm \frac{x}{4}$ or $y = 3 \pm \frac{x}{m}$ $2 \leqslant m \leqslant 6$ **OR** Sets the given equation of the curve $= ax + b$ and solves for a and b to find as a minimum to give as a minimum $y = k \pm \frac{x}{4}$ or $y = 3 \pm \frac{x}{m}$ $2 \leqslant m \leqslant 6$
A1	For the correct straight line.
M1	Draws their line correctly on the grid. Look for (0, 3) and (–4, 4) provided it is of the form $y = k \pm \frac{x}{4}$ You MUST see the equation of the line WITH the drawn line. A correct line without an equation is M0 
A1	For both $x = -0.2$ and $x = -3.3$ **Answers must be given to 1 dp only** [2 or more dp is A0] **NB:** Calculator values are -0.165, -3.323

Type MS · Solving an equation with your graph and a straight line

Answer 11 of 12 · Question 4 part whole · 4 marks

2024 JUN · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Solving an equation with your graph and a straight line

Answer 12 of 12 · Question 9 part c · 5 marks

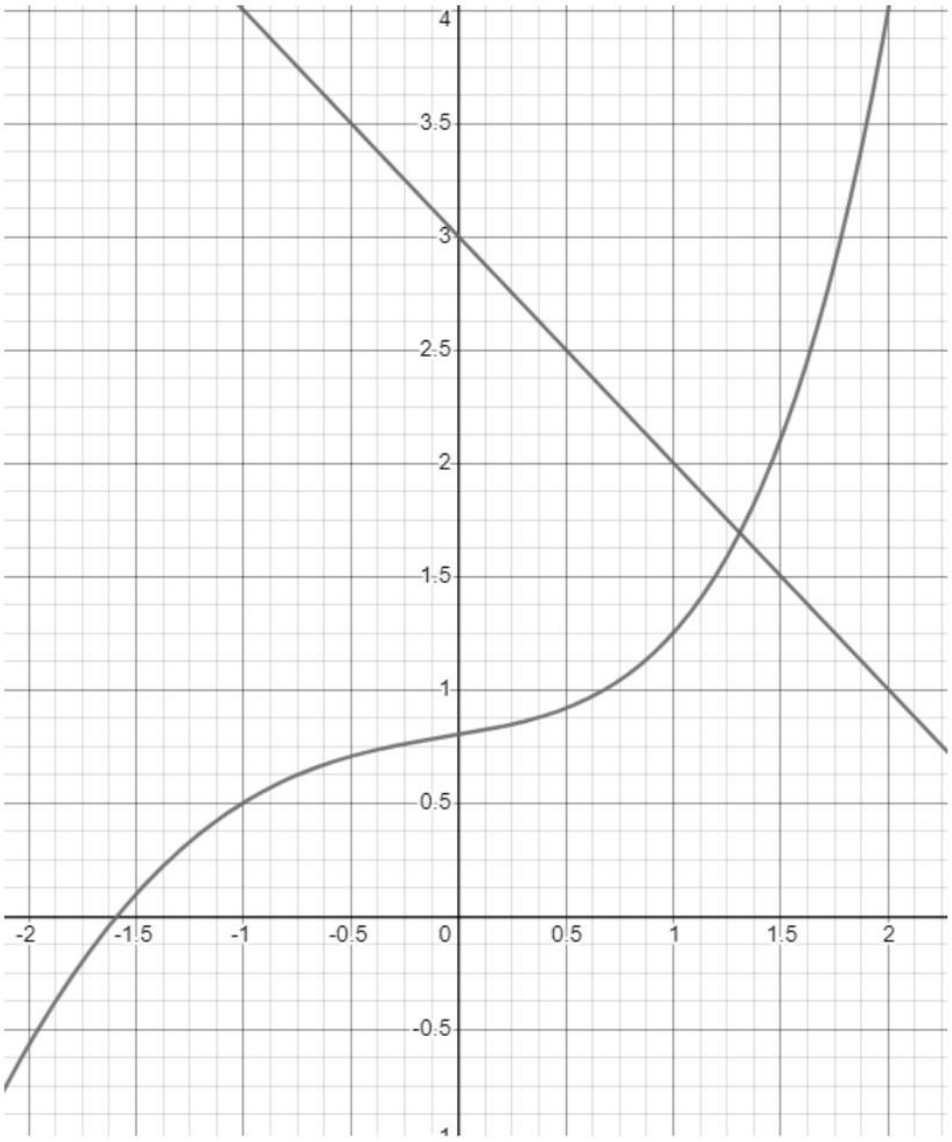
2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

9 (a)	$\begin{array}{\|c\|c\|c\|}\hline -0.5 & 0.5 & 1.5 \\ \hline 0.70 & 0.92 & 2.11 \\ \hline \end{array}$	B1 B1 [2]
(b)	Points plotted	B1ft
	Joined with a smooth curve	B1ft [2]
(c)	$x^3+4=x^2-8x+15$	M1
	$x^3+4=(x-5)(x-3)$ oe	M1
	$\dfrac{x^3+4}{5-x}=3-x$	A1
	$y=3-x$ drawn	M1
	$x=1.3/1.4$	A1 [5]
ALT	$\dfrac{x^3+4}{5-x}=ax+b \Leftrightarrow x^3+4=5ax-ax^2+5b-bx$	{M1}
	$\Leftrightarrow x^3+ax^2+\ b-5a\ \ x+4-5b\equiv x^3-x^2+8x-11$ $\Leftrightarrow a=-1\quad b=3$	{M1} {A1}
	$y=3-x$ drawn	{M1}
	$x=1.3/1.4$	{A1} [5]
		Total 9 marks

Type MS · Solving an equation with your graph and a straight line

Answer 12 of 12 · Question 9 part c · 5 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract



Type MS · Solving an equation with your graph and a straight line

Answer 12 of 12 · Question 9 part c · 5 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks